

MODELING THE EVOLUTION OF WESTERN POPULAR MUSIC SURPRISAL WITH IDYOM

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ABSTRACT

This paper investigates the historical evolution of melodic predictability in mainstream Western popular music using IDyOMpy, the latest Python implementation of the Information Dynamics of Music model. We analyze the BiM-MuDA dataset, which contains MIDI transcriptions of the top five singles from the Billboard Year-End Singles Charts for each year from 1950 to 2024, together with track metadata. After filtering the corpus to retain only valid monophonic melodies, we compute note-level surprisal and entropy values and aggregate them at the song level. To examine temporal change, we compare three evaluation settings: five-fold cross-validation over the full dataset, and rolling 5-year and 10-year look-back windows that model predictability relative to recent historical context. Results show a consistent downward trend in both surprisal and entropy, suggesting that mainstream melodies have become more predictable over time. This effect is stronger when chronological structure is explicitly modeled, highlighting the importance of temporal context for capturing long-term stylistic evolution.

1. INTRODUCTION

Music is a distinctive and characteristic human element. Throughout history, music has evolved alongside humanity, consistently serving as a powerful medium for communication, emotional expression, enjoyment, and social connection. Its evolution reflects the growth of human intellect and technological innovation. As our tools and knowledge have advanced, so too have our musical capabilities—leading to greater diversity in sound textures and a wider range of uses. More recently, the rise of streaming services has further expanded music’s accessibility, allowing people to explore and engage with it more easily than ever before. Nevertheless, there has been a notable trend that well-received and highly rated music has become increasingly generic, characterized by reduced melodic variety, similar

timbres, and homogenization of sound over the past few decades.

Within computational musicology, the study of musical evolution provides insight into broader cultural and cognitive processes. A central area of inquiry concerns the relationship between musical complexity and listener experience. In particular, music perception is closely tied to anticipatory processes, whereby listeners form expectations about forthcoming musical events. Prior work has demonstrated that measures of *surprisal* and *uncertainty* are strongly associated with perceived musical pleasure [1, 2].

Further studies have found that statistical methods for computing "surprisal" and "entropy" are significantly correlated with neural encoding of auditory stimuli [3]. In this context, IDyOM has become a widely adopted computational framework for modeling musical expectation. Its recent Python implementation enables scalable estimation of surprisal conditioned on a given musical corpus.

This work applies the IDyOM framework to the Billboard Melodic Music Dataset (BiMMDA) [4] to examine longitudinal changes in melodic surprisal and entropy from 1950 to 2024 in Western popular music. Results indicate a general decline in both measures over time, suggesting a reduction in melodic unpredictability. Genre-level analyses further reveal differentiated patterns of change, reflecting the evolving stylistic characteristics of popular music and varying degrees of convergence toward mainstream conventions.

2. RELATED WORK

Previous work has contributed to understanding the historical evolution of music from different perspectives. Some research focuses on analytical measures grounded in classic Digital Signal Processing (DSP) and Music Information Retrieval (MIR). These methods have proven useful not only for assessing the evolution of music [5], but also for offering cross-disciplinary insights into musical styles and their relationship to different cultural and historical periods [6]. Other methodologies focus on the evolution of the communicative and linguistic aspects of music [7], proving that the lyrics have become simpler and easier to comprehend and generalized emotion drifts towards negativity. In [8] musical compositions are encoded in weighted



networks to assess genre-related complexity, showing that Classical and Jazz compositions have higher complexity and melodic diversity than recently developed genres.

The idea that musical predictions and anticipation are key to music perception and cognition has been revisited and supported by several studies [9] [2] [10]. While musical structures explained by MIR metrics or exploratory data analysis over music representations offer valuable insights, there is a gap between these metrics and the user’s actual musical experience which in turn is related to the listener’s prior knowledge and expectations when listening to music. In this context, models such as IDyOM [11] reduce this gap by showing a high correlation between their numerical predictions and EEG data [3] [1], allowing us to use them as indicators of neural activity. From this perspective, the extracted features offer an analysis more closely related to human perception and cognition of music, rather than strict compositional and musical theory related metrics.

Although IDyOM’s measures of surprise and expectation are widely recognized as meaningful descriptors of musical perception and cognition [12], no prior work has specifically examined how these metrics evolve over time in Western mainstream music. This gap motivates the present study, which aims to investigate the historical development of these perceptual features. In doing so, this work contributes to a deeper musicological understanding of how Western mainstream music has evolved.

3. METHODOLOGY

To assess the evolution of predictability in mainstream Western music, this study employs the latest Python implementation of the IDyOM model [11]. The model is used to analyze melodic and rhythmic predictability, as well as variation, in musical compositions over time. It is applied to the **BiMMuDA** dataset [4], producing, for each song, average *surprisal* and *entropy* values as the final output.

3.1 BiMMuDA dataset and corpus preparation

The BiMMuDA dataset [4] contains 1170 MIDI files for the top five singles from the Billboard Year-End Singles Charts for each year from 1950 to 2024. It also includes metadata related to year and multi-label genre tagging. These annotations make the dataset suitable for analyzing structure and predictability.

Since the model requires symbolic monophonic input, the dataset was filtered to retain only valid monophonic MIDI files, removing corrupted and polyphonic ones. This resulted in 1,163 files, each assigned a single genre label using the `genre-broad-1` field.

Before the main modeling stage, preliminary descriptive analysis were carried out by aggregating song-level statistics and inspecting their distribution across years and genres. Figure 1 shows a clear imbalance, with most of the tracks belonging to *Pop* and *Rock*. Despite unbalance, this reflects their dominance in Western mainstream music since the 1950s.

3.2 IDyOM modelling

Predictive modeling was conducted using **IDyOMpy** [11], a Python implementation of the IDyOM framework. IDyOM estimates the probability of upcoming musical events using variable-order Markov models, combining both long-term and short-term learning components. The long-term model (LTM) is trained on a corpus to capture stylistic regularities, while the short-term model (STM) adapts online to the local context of each sequence.

In this study, the model was configured using the `pitch` and `onset` viewpoints to predict note-level distributions. For each event, IDyOM computes a probability distribution over possible continuations, from which two standard information-theoretic measures are derived: *surprisal* (negative log-probability of the observed event) and *entropy* (uncertainty of the predictive distribution). These measures were computed at the note level and subsequently averaged at the song level to facilitate comparison across items.

Three evaluation procedures were employed. First, a five-fold cross-validation was performed over the full valid BiMMuDA dataset, such that each fold was evaluated using models trained on the remaining data. This setting reflects a global stylistic model in which long-term knowledge is not temporally constrained.

To examine temporal dynamics, two additional diachronic evaluation schemes were implemented using rolling windows of five and ten years. In these conditions, the LTM was trained exclusively on data from preceding years and evaluated on the following year, thereby approximating historically informed expectations. The STM component operated as standard within each sequence. Model outputs were aggregated by song, year, and genre to support analyses of temporal trends and stylistic variation.

4. RESULTS

This section summarizes the results obtained using the methodology described above. Figure 2 presents the mean surprisal and entropy values per year, computed using the full dataset during the training stage. Figures 3 and 4 show the corresponding results for the 5-year and 10-year rolling-average models, respectively. All plots display both surprisal and entropy values, with green and red dotted line indicating the linear trend (slopes) fitted to each series. Table 1 summarizes the slope values for the different evaluation periods. Additionally, t-tests were conducted to assess whether the differences between the cross-validation series and the rolling-window series are statistically significant. The resulting p-values indicate that the null hypothesis of statistical equivalence can be rejected.

Tables 3, 4, and 5 present the slope values of the series computed for each genre. Genres with fewer than 10 testing data points were excluded from this analysis, as such small sample sizes are insufficient to provide reliable or representative results.

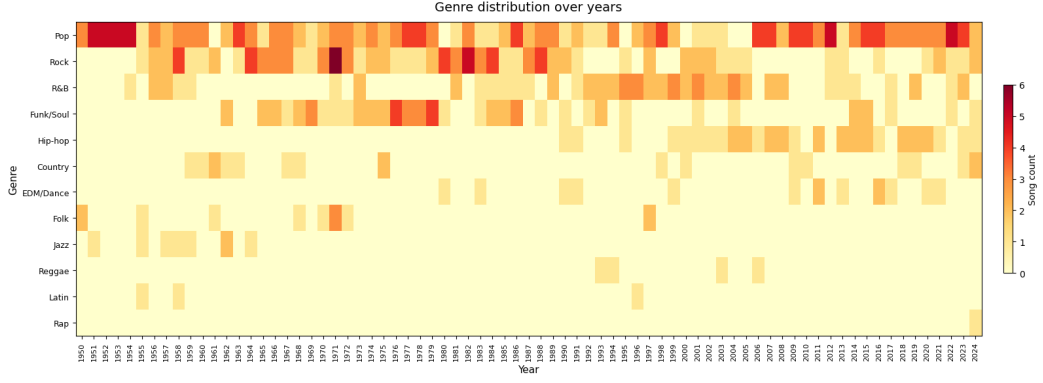


Figure 1: Genre distribution of songs throughout the years present in the BiMMuDA dataset. The presence of certain genres throughout the years informs how the rolling look-back window models evaluated surprisal.

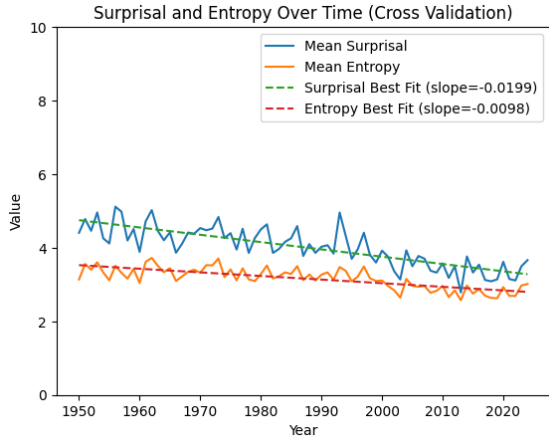


Figure 2: Mean surprisal and entropy over time using cross-validation method.

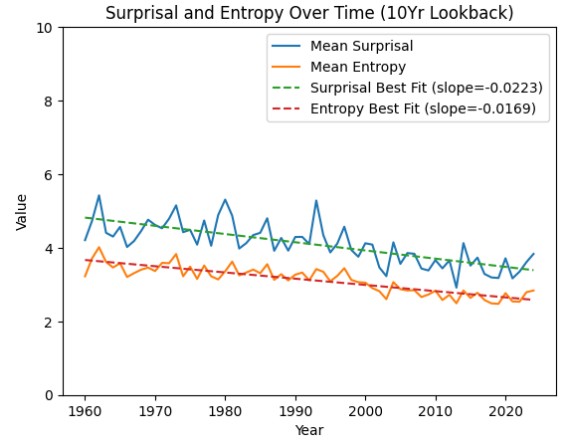


Figure 4: Mean surprisal and entropy over time using a 10-year rolling look-back window.

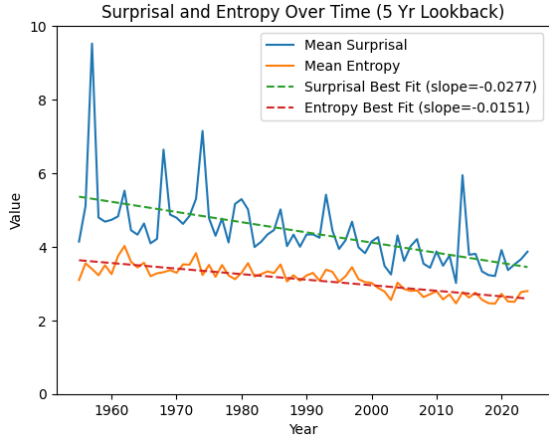


Figure 3: Mean surprisal and entropy over time using a 5-year rolling look-back window.

Evaluation Period	Entropy	Surprisal
Cross Val	-0.001	-0.020
5yr look-back	-0.015	-0.028
10yr look-back	-0.017	-0.022

Table 1: Linear trend slopes for surprisal and entropy across evaluation periods.

5. DISCUSSION

The presented results reveal a consistent downward trend in both surprisal and entropy across all evaluation settings, suggesting a gradual reduction in musical unexpectedness or unpredictability over time. This pattern is evident in the cross-validation setup and becomes more pronounced when temporal context is incorporated through rolling-window models.

A key finding is that models accounting for chronological structure (5-year and 10-year look-back) produce systematically steeper negative slopes than the cross-validation approach, with this difference being statistically significant (Table 2), particularly for the 5-year model ($p < 0.001$). This suggests that ignoring time can obscure how trends influence musical anticipation and highlights the importance of considering chronology in modeling musical enculturation and its influence on perceived surprisal and predictability. Furthermore, comparing the two temporal models shows that the 5-year window yields steeper slopes than the 10-year window for both surprisal and entropy, indicating greater sensitivity to local stylistic fluctuations. This likely reflects overfitting in the shorter window, leading to sharper probability distributions, lower entropy, and higher surprisal. In contrast, the 10-year model captures greater musical variability, resulting in higher entropy and more moderate surprisal changes. Overall, these

Comparison	Entropy	Surprisal
5yr vs CVal	$\approx 0^{***}$	$\approx 0^{***}$
10yr vs CVal	0.0435*	0.0329*

Table 2: t-test comparisons of surprisal and entropy trends across evaluation periods.

Genre	Entropy	Surprisal
Funk/Soul	-0.0091	-0.0132
Hip-hop	-0.0070	-0.0193
Pop	-0.0112	-0.0229
R&B	-0.0041	-0.0144
Rock	-0.0026	-0.0066

Table 3: Cross-validation: trend slopes by genre.

results point to a trade-off between sensitivity to short-term stylistic change and robustness to variability when selecting the temporal scope of training data in IDyOM.

The genre-based analysis reveals that the downward trend is not uniform across musical categories. Pop exhibits the most pronounced decline in both surprisal and entropy, indicating a stronger shift toward more predictable structures over time. Funk/Soul and Country also show notable decreases, particularly in longer temporal windows. In contrast, Rock displays relatively shallow slopes, suggesting greater stability in its structural characteristics. Interestingly, in the 5-year look-back setting, some genres such as R&B and Rock show slight increases in surprisal, pointing to short-term variability or localized increases in complexity. However, these effects are not observed in the 10-year model, reinforcing the interpretation that they reflect transient fluctuations rather than sustained trends, and that such trends are short-lived.

Several outliers with exceptionally high surprisal values were observed, particularly within the 5-year look-back condition. Closer inspection revealed that the three highest-surprisal cases in this setting correspond to instrumental melodies or instrumental sections of songs. This pattern suggests that structural differences in instrumental music, relative to predominantly vocal-based training data, may contribute to reduced model predictability.

In contrast, the lowest surprisal values (i.e., most predictable sequences) were consistently associated with EDM/Dance and Hip-hop genres. Notably, these low-surprisal songs remained consistent across all evaluation conditions, indicating a systematic degree of predictability within these styles.

6. CONCLUSIONS AND FUTURE WORK

Overall, the results support the hypothesis of a long-term trend toward reduced melodic complexity, as reflected in declining surprisal and entropy, with additional variation across genres and temporal scales. Comparisons across cross-validation and rolling-window conditions indicate that temporal context systematically influences estimated surprisal, contributing to the appearance of increasing homogeneity over time.

Notably, the steeper decline observed in the 5-year

Genre	Entropy	Surprisal
Pop	-0.0073	-0.0068
R&B	-0.0024	0.0011
Rock	-0.0021	0.0050

Table 4: 5-year look-back: trend slopes by genre.

Genre	Entropy	Surprisal
Country	-0.0141	-0.0167
Funk/Soul	-0.0146	-0.0178
Pop	-0.0181	-0.0240
Rock	-0.0076	-0.0067

Table 5: 10-year look-back: trend slopes by genre.

look-back condition suggests that shorter training windows constrain the diversity of the learned distributions. This limitation likely affects model calibration, leading to less reliable probability estimates and, consequently, surprisal values.

Despite the consistency of the observed trends, several limitations should be acknowledged. First, the analysis relies on the IDyOM framework, which models expectation from symbolic sequences. While effective for capturing tonal and sequential structure, it does not account for other dimensions such as timbre, dynamics, production, or expressive performance. Therefore, the observed decrease in *surprisal* and *entropy* should be interpreted as reduced predictability in the symbolic domain, not overall musical complexity.

Second, the use of BiMMuDA introduces potential biases. Although BiMMuDA facilitates large-scale modeling and comparison across a wide range of years, it is unevenly distributed across genres and limited to Western mainstream music. This may affect both the estimated slopes and the apparent strength of temporal trends. Additionally, the exclusion of genres with fewer than 10 data points, while necessary for statistical reliability, further limits the generalization of the genre-specific findings.

Finally, the temporal modeling approach has limitations. Rolling windows (5-year and 10-year) impose fixed window sizes that may not align with the natural timescales of musical change. Moreover, the assumption of linear trends can oversimplify potentially nonlinear or period-specific dynamics in the evolution of musical structure.

Future work could address these limitations in several ways. One promising direction is to extend the analysis beyond symbolic representations by incorporating audio-based features, allowing for a more holistic characterization of musical complexity. From a data perspective, expanding and re-balancing the dataset across genres, regions, and time periods would improve the robustness of the findings. More fine-grained genre classifications or sub-genre analysis could also reveal patterns that are not captured at the current level of aggregation. Furthermore, exploring adaptive or variable-length temporal windows, as well as nonlinear trend modeling, could provide a more nuanced understanding of how musical characteristics evolve over time.

7. SOURCE CODE

The source code and supplementary analysis for this project is open-source and available for download on GitHub at the following link: https://github.com/martiinssssss/IDyOM_TimeMachine

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